DERIVATIVES 101 · DECK 2

Reverse convertibles

Equity and FX. Plus the callable, autocallable and lock-in variants that make up most of what actually trades.

Structured products onboarding · Development team · August 2026

IN ONE SENTENCE

A high coupon, paid for with your principal

A note that pays an above-market coupon, and repays your money in full only if the underlying stays above a level. Below it, you get the asset instead of your cash.

It is 'reverse' because a normal convertible lets the investor choose to convert. Here the issuer decides — and only ever converts when it hurts.

Deposit or bond

Repays 100 at maturity. This is the wrapper.

+ short put

Sold by the client, struck at or below spot. This is the risk.

= high coupon

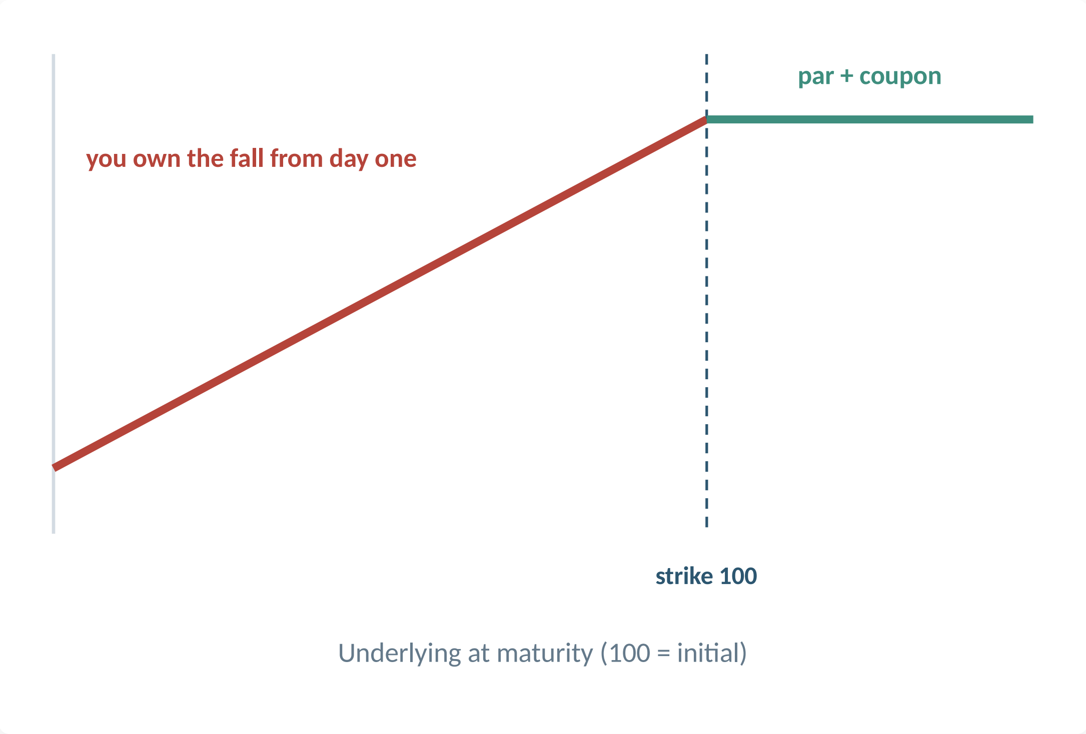
The put premium, spread over the life of the note.

If you understood the short put in Deck 0, you already understand this product. Everything from here is packaging.

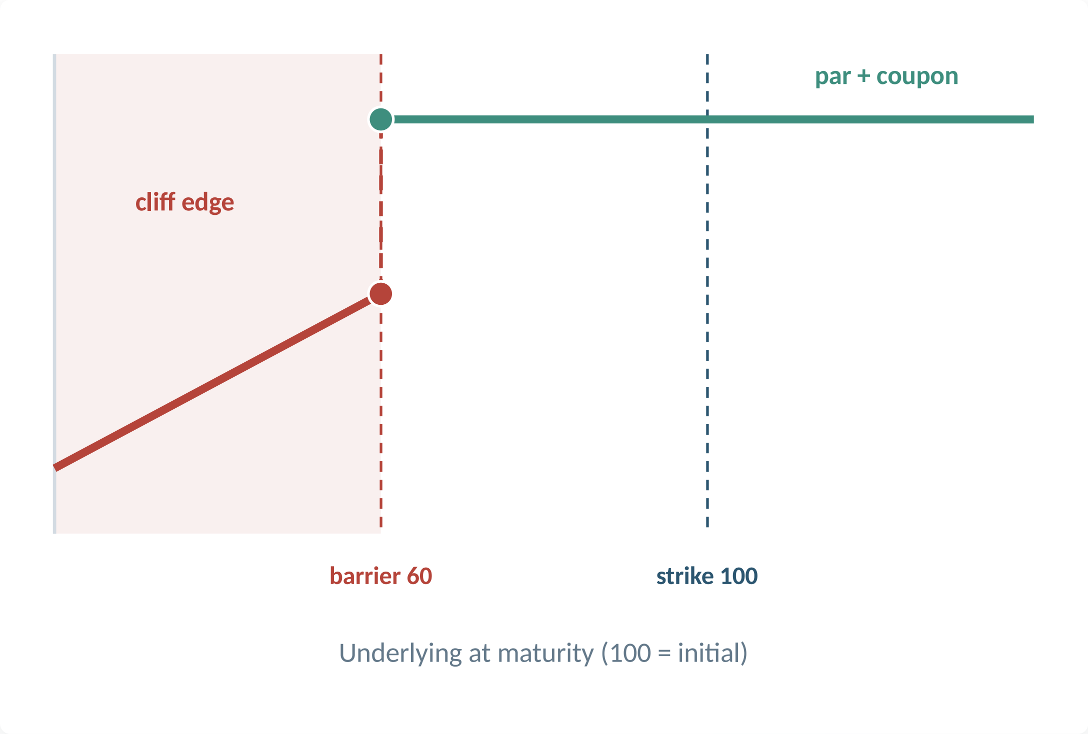
THE TWO SHAPES

Adding a barrier changes the picture completely

Vanilla reverse convertible — strike at 100



Barrier reverse convertible — 60% knock-in



The barrier does not reduce the loss — it postpones it. One point below the barrier the client loses 40 points, instantly.

THE FORK

Is the coupon guaranteed, or does it have to be earned?

SSPA 1230

Unconditional

The coupon is paid on every payment date whatever the underlying does. The barrier affects redemption of principal only. This is the plain barrier reverse convertible, and it is still the Swiss convention.

SSPA 1255 / 1260

Conditional

Each coupon is paid only if the underlying is above a coupon barrier on that observation date. The client has sold an extra strip of digital options on top of the put.

- **The distinction is worth more than it looks.**
- A conditional coupon can stop precisely when the client most needs the income — in a drawdown — and the headline rate on the cover is the rate they will not be receiving.
- It is also a different product for us: an unconditional coupon is a fixed schedule, a conditional one is a schedule of contingent cashflows that has to be evaluated at every observation.
- Memory features exist to soften this. They are close to standard in Italy, common in Switzerland, and an optional extra in the US.

A conditional-coupon product is usually not booked under 1230. Check the category code as well as the name — they are set independently and both can be wrong.

TWO BARRIERS

The coupon barrier and the knock-in barrier price different risks

Coupon barrier

Observed on each coupon date. Decides whether that period's coupon is paid. Missing it costs income, not principal.

Knock-in barrier

Decides whether principal is repaid at par or the client takes the underlying's loss. Usually observed at maturity, sometimes continuously.

Same structure, two settings

Both at the same level

One number for the client to watch. The common case.

Payout Threshold below the Barrier

Coupons keep paying after principal protection has already gone.

Coupon barrier above the Barrier

Income stops long before principal is at risk. Buys a bigger headline.

- **Setting both at the same level is the dominant convention — one number for the client to watch.**
- They can sit either way round, and which way tells you what the structurer was optimising — a lower Payout Threshold protects the income stream, a higher one buys a bigger number on the cover.
- Check the observation convention separately for each barrier. Protection is often observed continuously while the coupon is tested once per period, at the same level.

A live Swiss note carries a continuously observed Barrier at 80% for principal and a Payout Threshold at 60% for the coupon — the second arrangement above, and the one that surprises people.

THE NAMING PROBLEM

One payoff, five names, and a code that means two things

Switzerland	Conditional Coupon Reverse Convertible, with or without a barrier	SSPA 1255 / 1260
United States	Contingent Income Auto-Callable, or Phoenix Autocallable	No formal category
Italy and France	Phoenix — as against Athena, which autocalls but pays no periodic coupon	EUSIPA 1260, Barrier Express
Germany	Folded into the Express-Zertifikat family	EUSIPA 1260
Korea	Step-down autocallable ELS	Statutory class, not a marketing name

The code 1260 means a conditional-coupon barrier reverse convertible in Switzerland and an express certificate everywhere else in Europe.

Same number, different product. On our own paperwork the two Swiss codes are used precisely: 1255 for a conditional coupon with no barrier, 1260 once a barrier is added.

WORKED EXAMPLE

A live US autocallable, term by term

Issuer	A bank; the client takes its credit risk for three years
Underlyings	Worst-of three indices — Russell 2000, Nasdaq-100, Nikkei 225
Tenor	3 years, with monthly observations
Coupon	0.958% per month ≈ 11.50% p.a., contingent
Coupon barrier	60% of each index's initial level
Autocall trigger	100% of initial, flat — no step-down
Non-call period	First call observation is roughly three months in
Downside threshold	60%, observed only at maturity (European)
At maturity	Above 60%: par. Below 60%: par × worst performer, so a 45% index means a 55% loss

The three questions

What did the client sell? A down-and-in put on the worst of three indices.

What makes it worse? Any one index falling 40%, and the three of them decoupling.

When does it end? Most likely at the first observation after a rally — not in three years.

11.50%

coupon p.a., contingent

−40%

before principal is lost

Terms from a Nomura Holdings 424B2 filing, issued 1 May 2026, US$7.1m. Representative rather than exceptional.

WHAT EACH FEATURE COSTS

One share, one morning, nine structures

Twelve-month notes on the same share, priced the same day. Where there is a barrier it is at 80%. Only one thing changes at a time. Levels are indicative and rounded to a tenth of a point.

Reverse convertible, no barrier	11.6%	The base case. Principal is at risk from the first point below the strike.
+ continuous 80% barrier	10.7%	The barrier costs about nine tenths of a point. That is what conditional protection is worth.
barrier on the daily close instead	10.6%	Softer observation, so the barrier survives more often — and the client pays for it in coupon.
+ quarterly autocall at 100%	10.9%	Pays more than the plain barrier note: the client has sold the early-redemption optionality.
autocall stepping down 100 / 99 / 98 / 97	10.7%	Easier to redeem, so the expected life is shorter and the quoted rate eases back.
callable at the issuer's discretion	11.7%	Roughly a full point more than the mechanical autocall. Discretion is the most expensive feature here.
callable, quanto into a second currency	7.9%	Removing the currency risk costs nearly four points. Quanto is never free.
conditional coupon, no barrier · 1255	11.9%	The coupon is no longer guaranteed, so the headline rises.
conditional coupon with barrier · 1260	11.3%	Both features together. Compare it with the 11.6% base case and you can price each one.

These are all the same trade on the same share. The coupon differences are a price list for the features.

FLAVOUR 1

Callable: the issuer decides, and it is not doing you a favour

- **The issuer holds an option to redeem the note early, at par plus accrued coupon.**
- It exercises when the note has become expensive for it to keep — typically when volatility has fallen or the underlying has rallied.
- That is precisely when the client would rather keep receiving the coupon. Optionality that belongs to the issuer is optionality the client has sold.
- In exchange the client gets a higher headline coupon than the equivalent non-callable note. That gap is the price of the call.
- **Daily-callable variants exist in Hong Kong retail ELIs. The issuer can call on any business day, which makes the expected life very short.**

Callable

Issuer's discretion. Exercised in the issuer's interest, on its own model.

Autocallable

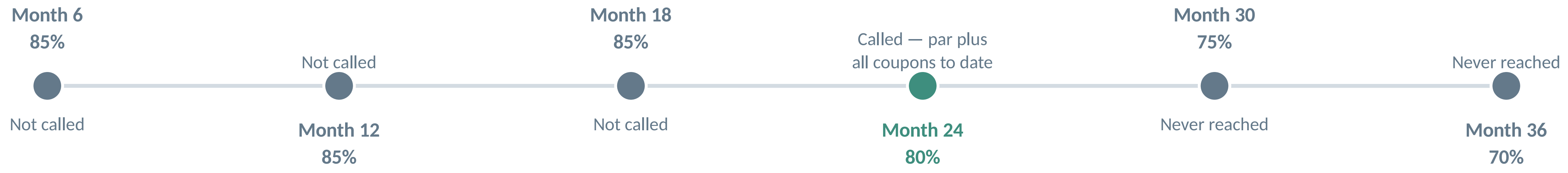
Mechanical. Triggered by an observable level, with no discretion on either side.

Clients almost always prefer autocallable, because 'the bank decides' is a hard sentence to put in a suitability file.

FLAVOUR 2

Autocallable: the structure that dominates the market

A step-down schedule makes the note progressively easier to redeem



Why it steps down

Each observation calls at or below the level of the last. The issuer wants the note gone; the client wants coupons. The schedule is a negotiated compromise.

Why the desk likes it

Rebalancing risk is spread across six trigger points instead of concentrated at one. Post-2015 'lizard' barriers do the same job.

Why it matters to us

A single trade generates six conditional cashflow branches. Get the observation calendar, business-day conventions and non-call window wrong and every downstream number is wrong.

Schedule shown follows a real Korean single-stock ELS: 85 / 85 / 85 / 80 / 75 / 70, semi-annual observations, three-year tenor.

FLAVOUR 3

Lock-in and memory: coupons that are deferred, not lost

Memory coupon — 1.0% per month, barrier at 80%

M1	M2	M3	M4	M5	M6
above barrier	above barrier	below barrier	below barrier	above barrier	above barrier
1.0%	1.0%	—	—	3.0%	1.0%

months 3 and 4 are remembered and paid in month 5

Memory coupon

In the documentation this is a Payout Postponement Event: the coupon is not lost, it is postponed and added to the next Payout Amount. Also called a snowball coupon.

Lock-in (bonus / capped)

Once the underlying trades above a level, a minimum return is banked for good and cannot be lost, whatever happens afterwards.

Why it sells

It removes the single most painful client conversation — 'you missed one observation by 0.3% and lost the coupon' — for a modest reduction in headline rate.

Worst-of: paying the client to sell dispersion

- **Adding a name to the basket raises the coupon because it lowers the expected payout.**
- The note references the weakest performer. A fourth name can only make the weakest performer weaker — never stronger.
- The worst-of option is worth most when the names decouple, because the worst performer is then further from the pack. The client has sold that dispersion; the desk, which now owns the option, is the side that is short correlation.
- That matters because correlation rises in a crash — the desk's position deteriorates exactly when the market is moving fastest. Desks recycle the exposure through correlation and covariance swaps.
- **Two to four names is the norm. Three is the most common. Every additional name is more coupon and materially more risk.**

A three-name worst-of at maturity

Nvidia	+42%
Microsoft	+9%
Tesla	−48%

Barrier at 60%

Tesla at 52% of initial is through the barrier.
The note redeems at 52% of par.
The other two names are irrelevant.

THE FX COUSIN

Dual currency investments: the same trade in FX

Structure	Deposit in a base currency + short FX put. The premium is paid out as enhanced interest
At maturity	If the alternate currency has not weakened through the strike: repaid in base currency with the enhanced rate
Otherwise	Principal converted at the strike rate into the weaker currency — so the client is long the currency that just fell
Tenor	1 week to 6 months. Far shorter than equity structures
Pairs	Any combination of HKD, USD, CNY, AUD, NZD, GBP, CAD, EUR, JPY
Minimum	Around HK$50,000 equivalent at a Hong Kong retail bank
Principal protection	None, in base-currency terms. Banks state this explicitly on the product page

Two fields, side by side

A one-month FX note shows Coupon 11.7% p.a. and, three lines below it, Maximum Yield 1.0%. Both are correct. Always read the second number.

Where it goes wrong

Clients roll these repeatedly, collecting small premiums, until one conversion leaves them holding a currency they did not want at a rate they cannot recover from. The Taiwanese target-redemption forward crisis was this pattern with leverage bolted on.

Terms illustrative of live Hong Kong retail dual currency investment offerings.

WHAT ACTUALLY TRADES

Typical terms, and how much they vary by market

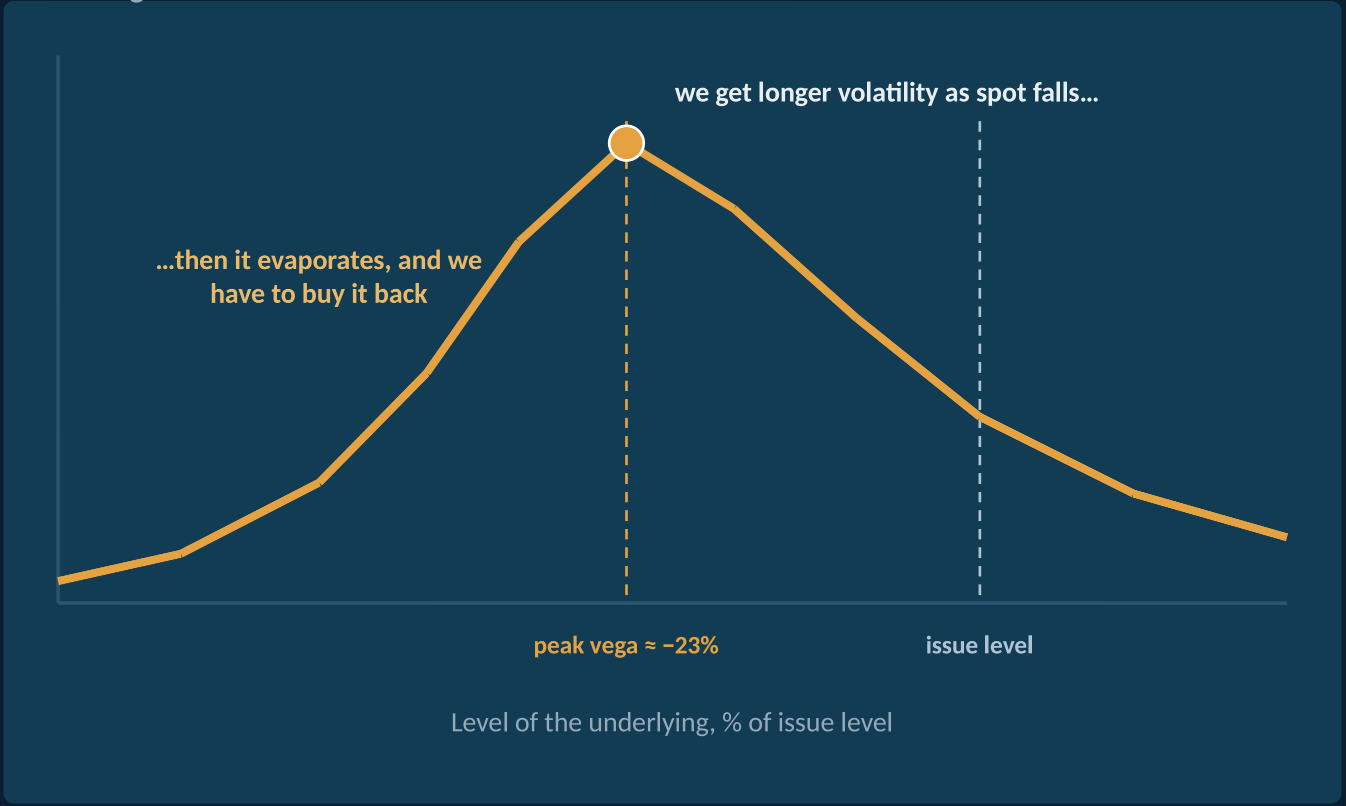
Tenor	1–3 years in the US and Europe. 3 years standard in Korea. Under 1 year for 98% of Taiwanese trades
Coupon	Roughly 8–12% p.a. on an index worst-of; 20%+ on a volatile single-stock worst-of
Knock-in barrier	60–65% US and Europe. 50% typical for Korean index ELS. 35–45% on high-volatility single stocks
Observation of barrier	Increasingly European — observed only at maturity. A real de-risking versus the continuous barriers of the 2015 vintage
Autocall trigger	100% flat is common in the US and Switzerland; gentle Swiss step-downs run 100/99/98/97, Korean ELS 85/85/85/80/75/70
Observation frequency	Monthly or quarterly for index notes; semi-annual for Korean single-stock ELS
Non-call period	Three to six months from trade date
Basket size	2–4 names, three being the modal US structure

Composite of published 2025–26 term sheets and market commentary. Always check current levels against the desk's own trade tape before quoting.

THE DESK'S PROBLEM

Peak vega: the hedge that turns against you on the way down

Dealer vega on an autocallable book



The flip

As spot falls, autocall probability collapses, expected life extends, and the book gets longer vega — until the barrier comes into view and the embedded put becomes near-certain. Then vega falls away sharply.

The crowd

Every dealer issues on the same handful of indices at similar levels in the same vintage. The flip happens to all of them at once, in the same direction, in a market with no liquidity.

The precedents

Korea 2015 (≈US$300m of street losses), Natixis Q4 2018 (€260m), and March 2020, when European dividend cancellations added an unhedged loss on top of peak vega.

Curve shown is a schematic of the shape the desk describes, not a calibrated model output.

THE FEEDBACK LOOP

Autocall books supply the market's long-dated volatility

1

Volatility is high

Coupons look attractive. Clients buy.

2

Issuance rises

Dealers take on the other side and are net long vega.

3

Dealers sell volatility

Hedging that vega suppresses implied volatility, especially at long tenors.

4

Volatility falls

Coupons look less attractive. Issuance slows, and the suppression unwinds.

When Japan and Korea curbed retail structured product sales, long-dated Nikkei and HSCEI options lost their main source of supply and implied volatility rose.

This is why the desk cares about regulation in markets it does not sell into. Issuance in one jurisdiction sets the price of volatility for everyone. Hong Kong is now a larger supplier of structured-product vega than Korea or Japan.

WHEN IT GOES WRONG

Korea 2024: the largest retail loss on record

KRW 19.3tn

HSCEI-linked ELS outstanding

As at November 2023; 82% sold through five commercial banks

–58%

HSCEI drawdown from strike

From above 12,000 in 2021 to about 5,100 in early 2024

KRW 4.6tn

realised investor losses

Roughly US$3.2bn, concentrated in the first half of 2024

20–65%

compensation ratios

Set by the FSS in March and May 2024, varying by bank

- **The product did what it said it would. The distribution did not.**
- Investors bought in 2021 with the index near its peak. Three-year notes matured in 2024 after the index had roughly halved — well through 50–65% knock-in barriers.
- The regulator's finding was about sales, not structure: 90.6% of bank sales were in-branch, and ELS counters were not separated from deposit counters, so clients conflated the two.
- Remedies: bank ELS sales restricted to 5–10% of branches with physically separated areas; sales resumed September 2025; a fine of roughly KRW 600bn across five banks was reported in June 2026, still subject to approval.
- From August 2026, a mandatory alert fires when an underlying comes within 10 percentage points of the knock-in barrier.

WHAT TO TAKE AWAY

Five things about reverse convertibles

1. It is a bond plus a short put. The coupon is the put premium, spread out over time.
2. A barrier postpones the loss, it does not shrink it. One point through the barrier costs 40 points.
3. Callable means the issuer chooses; autocallable means a level chooses. Clients should know which they own.
4. Every extra name in a worst-of raises the coupon and lowers the expected payout. That is not a coincidence.
5. The desk's risk is not the mirror of the client's. Peak vega, correlation and dividends fail on their own schedule.

Next: Deck 3 — the Asian payoffs. Fixed coupon notes and equity linked notes.